

BRIEF REPORT

Serum Retinol-Binding Protein-4, Leptin, and Adiponectin Concentrations Are Related to Ectopic Fat Accumulation

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Context: Serum retinol-binding protein 4 (RBP-4), leptin, and adiponectin concentrations identify insulin resistance in varied conditions, but their relationships with insulin sensitivity and ectopic fat accumulation are unclear.

Objective: Our objective was to establish how these adipokines are related with intramyocellular lipid (IMCL) and intrahepatic lipid (IHL) content.

Design and Setting: We assessed retrospectively serum fasting RBP-4 concentrations in 1) 53 nondiabetic individuals in which insulin sensitivity and IMCL content were assessed by means of the insulin clamp and of ^1H magnetic resonance spectroscopy of the calf muscles, and 2) 140 nondiabetic individuals in which insulin sensitivity and the IHL content were assessed by means of the updated homeostasis model assessment and of ^1H magnetic resonance spectroscopy. In both experiments, serum leptin and adiponectin concentrations were measured.

Results: Fasting serum RBP-4, adiponectin, and leptin were associated with peripheral insulin sensitivity, were abnormal in the first-degree relatives of type 2 diabetic parents, and correlated with the soleus IMCL content and with the IHL content. The association of RBP-4 and adiponectin with insulin sensitivity was age, sex, and body mass index independent, but stepwise regression analysis suggested that RBP-4, but not adiponectin and leptin, was independently associated with insulin sensitivity. Adiponectin was independently associated with the IHL content, RBP-4, and leptin with the soleus IMCL content.

Conclusion: Serum RBP-4 was a robust marker of insulin resistance. Serum RBP-4, leptin, and adiponectin concentrations reflected ectopic fat accumulation in humans. (*J Clin Endocrinol Metab* 92: 4883–4888, 2007)

SEVERAL ADIPOSE-DERIVED PEPTIDES have been suggested to act as a potential link between accumulated fat mass and altered insulin sensitivity (1). The serum levels of retinol-binding protein 4 (RBP-4) were found to be increased in insulin-resistant individuals (2, 3), and it was also suggested that serum RBP-4 may be a noninvasive and accessible method reflecting the risks of impaired glucose tolerance, type 2 diabetes, and cardiovascular disease (3). Because insulin resistance has been shown to be tightly associated with this adipokine, we hypothesized that increased serum RBP-4 may be a feature of nondiabetic individuals with excessive ectopic fat accumulation. We tested this hypothesis in a cross-sectional fashion using noninvasive state-

of-the-art assessment of the intramyocellular lipid (IMCL) and intrahepatic lipid (IHL) content by means of ^1H magnetic resonance spectroscopy (^1H -MRS). Moreover, the relationships with adiponectin and leptin (1), also believed to be involved in the regulation of insulin sensitivity and body fat distribution, were analyzed.

Subjects and Methods

Serum RBP-4 concentration was measured retrospectively in fractions of spared serum samples from individuals who participated in two previously reported studies.

Study 1

The first study explored the relationships between insulin sensitivity and IMCL content in offspring of type 2 diabetic parents and controls (4). Basically, subjects were studied by means of 1) the euglycemic-hyperinsulinemic clamp to assess whole-body insulin sensitivity, 2) ^1H -MRS of the calf muscles to assess soleus and tibialis anterior IMCL content, and 3) dual x-ray absorptiometry for the assessment of body composition. Appropriately stored fractions of spared serum for the assessment of RBP-4 were available for 53 individuals, 15 offspring of type 2 diabetic parents, and 38 controls. Their characteristics are summarized in Table 1.

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Abbreviations: BMI, Body mass index; CV, coefficient(s) of variation; FFA, free fatty acids; HDL, high-density lipoprotein; HOMA2, updated computer homeostasis model assessment; IHL, intrahepatic lipid; IMCL, intramyocellular lipid; MRS, magnetic resonance spectroscopy; RBP-4, retinol-binding protein 4; $\text{SI}_{\text{P(clamp)}}$, insulin sensitivity; ww, wet weight.

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TABLE 1. Anthropometric, laboratory, and metabolic characteristics of the 53 nondiabetic individuals of protocol 1 segregated by family history of type 2 diabetes

	Offspring	Normal
n (female/male)	10/5	22/16
Age (yr)	31 ± 9	26 ± 5
BMI (kg/m ²)	24.6 ± 7.6	24.2 ± 5.7
Physical activity index	8.5 ± 1.2	8.2 ± 1.7
Total cholesterol (mmol/liter)	5.04 ± 0.85	4.50 ± 0.67
HDL cholesterol (mmol/liter)	1.06 ± 0.34	1.11 ± 0.36
Triglycerides (mmol/liter)	1.24 ± 0.70	0.87 ± 0.46
Serum FFA (mmol/liter)	0.66 ± 0.15 ^a	0.50 ± 0.22
Plasma glycerol (mmol/liter)	36.7 ± 17.5	44.3 ± 23.9
Plasma glucose (mmol/liter)	5.27 ± 0.44	5.05 ± 0.22
Serum insulin (pmol/liter)	45 ± 16	39 ± 19
Soleus IMCL (AU)	94 ± 39 ^b	67 ± 33
Tibialis anterior IMCL (AU)	31 ± 20	15 ± 10
SI _{P(clamp)} [liters/min·kg LBM/(pmol/liter) × 10 ³]	12.0 ± 4.1 ^a	17.2 ± 6.1
Adiponectin (μg/ml)	9.4 ± 4.5 ^b	12.8 ± 6.1
Leptin (ng/ml)	8.9 ± 8.5	9.0 ± 7.9
RBP-4 (μg/ml)	28.9 ± 8.2 ^b	23.2 ± 5.7

IMCL content was assessed using ¹H-MRS. Physical activity index is a score of habitual physical activity based on three meaningful factors: 1) occupational physical activity (precoded according to three levels of physical activity at work); 2) sport during leisure time calculated from a combination of the intensity, frequency, and proportion of the year during which the sport was played regularly; and 3) other physical activity during leisure time (this specifically relates to watching television, walking, and cycling during leisure time). AU, Arbitrary units; LBM, lean body mass.

^a *P* < 0.01 vs. normal.

^b *P* < 0.05 vs. normal (two-tailed independent *t* test).

Study 2

The second study assessed the prevalence of fatty liver in a population of young, nondiabetic, healthy individuals and explored the impact of phenotypic variables, in particular habitual physical activity, on the IHL content (5). Basically, subjects were studied by means of ¹H-MRS of the liver to assess the IHL content. Appropriately stored fractions of spared serum for the assessment of RBP-4 were available for 140 individuals, 24 offspring of type 2 diabetic parents, and 116 controls. Their characteristics are summarized in Table 2 with individuals segregated

TABLE 2. Anthropometric, laboratory, and metabolic characteristics of the 140 nondiabetic individuals of protocol 2 segregated by fatty liver

	Fatty liver	Normal
n (female/male)	1/23 ^a	46/70
Age (yr)	34 ± 8	36 ± 7
BMI (kg/m ²)	28.1 ± 3.8 ^b	23.7 ± 3.2
Physical activity index	8.3 ± 1.4	7.9 ± 1.6
Total cholesterol (mmol/liter)	5.18 ± 1.35 ^b	4.60 ± 0.79
HDL cholesterol (mmol/liter)	1.17 ± 0.28 ^b	1.52 ± 0.38
Triglycerides (mmol/liter)	1.69 ± 1.31 ^b	0.86 ± 0.37
Serum FFA (mmol/liter)	0.61 ± 0.15	0.56 ± 0.23
Plasma glucose (mmol/liter)	5.13 ± 0.53	4.75 ± 0.54
Serum insulin (pmol/liter)	106 ± 36	71 ± 32
IHL content (%ww)	13.9 ± 8.5 ^b	1.7 ± 1.1
HOMA2-%S	50 ± 22 ^b	83 ± 47
HOMA2-%B	160 ± 41	147 ± 63
Adiponectin (μg/ml)	5.4 ± 2.1 ^b	8.3 ± 3.6
Leptin (ng/ml)	9.5 ± 4.8	8.6 ± 7.8
RBP-4 (μg/ml)	34.8 ± 7.5 ^b	30.5 ± 7.8

IHL content was assessed using ¹H-MRS. HOMA2-%S, HOMA insulin sensitivity; HOMA2-%B, HOMA β-cell sensitivity.

^a *P* < 0.05 (Pearson χ^2 test).

^b *P* < 0.01 vs. normal.

by IHL content of more than 5% wet weight (ww) (fatty liver) or less than 5% ww (normal). Because of its potential impact on insulin sensitivity (6), women of both study 1 and 2 were not taking oral steroidal contraception for at least 12 months and were studied between d 3 and 8 of the menstrual cycle. All recruited subjects gave their informed written consent after explanation of the purpose, nature, and potential risks of the study. The protocols were approved separately by the Ethical Committee of the San Raffaele Scientific Institute.

Euglycemic-hyperinsulinemic clamp

Briefly, in study 1, the clamp was performed at 0700 h after a 10-h overnight fast and was combined with a bolus (5 mg/kg body weight) plus continuous infusion (0.05 mg/kg body weight/min) of 6,6-[²H₂]glucose obtained from MassTrace (Woburn, MA). After a 150-min tracer equilibration period, a euglycemic-hyperinsulinemic clamp was performed using an insulin infusion rate of 40 mU/m²·min.

¹H-MRS

In study 1, soleus and tibialis anterior ¹H-MRS was performed on a GE Signa 1.5-T scanner (General Electric Medical Systems, Milwaukee, WI) as previously described (4, 6). In study 2, hepatic ¹H-MRS was performed in all volunteers with the use of a 1.5-T whole-body scanner (Gyrosan Intera Master 1.5 MR system; Philips Medical Systems, Best, The Netherlands) as previously described (5).

Analytical determinations

In both study 1 and 2, fasting plasma or serum glucose, free fatty acids (FFA), glycerol, triglycerides, total cholesterol, and high-density lipoprotein (HDL)-cholesterol were measured as previously described (4–6). Serum levels of insulin [sensitivity of 2 μU/ml and intra- and interassay coefficients of variation (CV) of <3.1 and 6%, respectively] and leptin (sensitivity of 0.5 ng/ml and intra- and interassay CV of <5 and 9%, respectively) were measured with RIA (Linco Research, St. Charles, MO). Serum adiponectin was measured by ELISA kit (B-Bridge International, Inc., Sunnyvale, CA) with a sensitivity of 25 pg/ml. The intraassay CV was less than 3.7% and interassay CV less than 6%. Serum RBP-4 (sensitivity of 0.9 ng/ml and intra- and interassay CV of <3 and 9.8%, respectively) was measured by means of an ELISA kit (catalog no. K6110; Immundiagnostik AG, Bensheim, Germany). The d₂-glucose enrichment was measured by gas chromatography/mass spectrometry as previously described (4, 6).

Calculations

In study 1, peripheral insulin sensitivity [SI_{P(clamp)}] was obtained as follows: $\Delta R_d / (\Delta I \times G)$, where ΔR_d is the increment of total glucose uptake (normalized to kilograms lean body mass), ΔI is the increment of serum insulin concentration (both calculated at basal and clamp steady-state conditions), and *G* is the plasma glucose concentration during the clamp (7). In study 2, insulin sensitivity was determined by the Updated computer homeostasis model assessment (HOMA2) indexes (5) available from www.OCDem.ox.ac.uk.

Statistical analysis

Data in text and tables are means ± sd. Analyses were performed using the SPSS software (version 13.0; SPSS Inc., Chicago, IL). Comparison between groups was performed using two-tailed independent-samples *t* test, and a *P* value of < 0.05 was considered to be significant. Variables with skewed distribution assessed using Kolmogorov-Smirnov test of normality [body mass index (BMI), IHL content, HDL-cholesterol, triglycerides, and glucose] were log-transformed before the analysis. The relationship between serum adipokines concentration and continuous variables were examined by two-tailed Pearson's correlation coefficients; partial correlation was used to examine these relationships independently of other variables. To ascertain how important the relative contribution of fasting adiponectin, leptin, and RBP-4 concentrations is to predict insulin sensitivity and IMCL and IHL content, we used stepwise regression analysis (using *F* ratio-to-remove of 4 and *F* ratio-to-enter of 3.996).

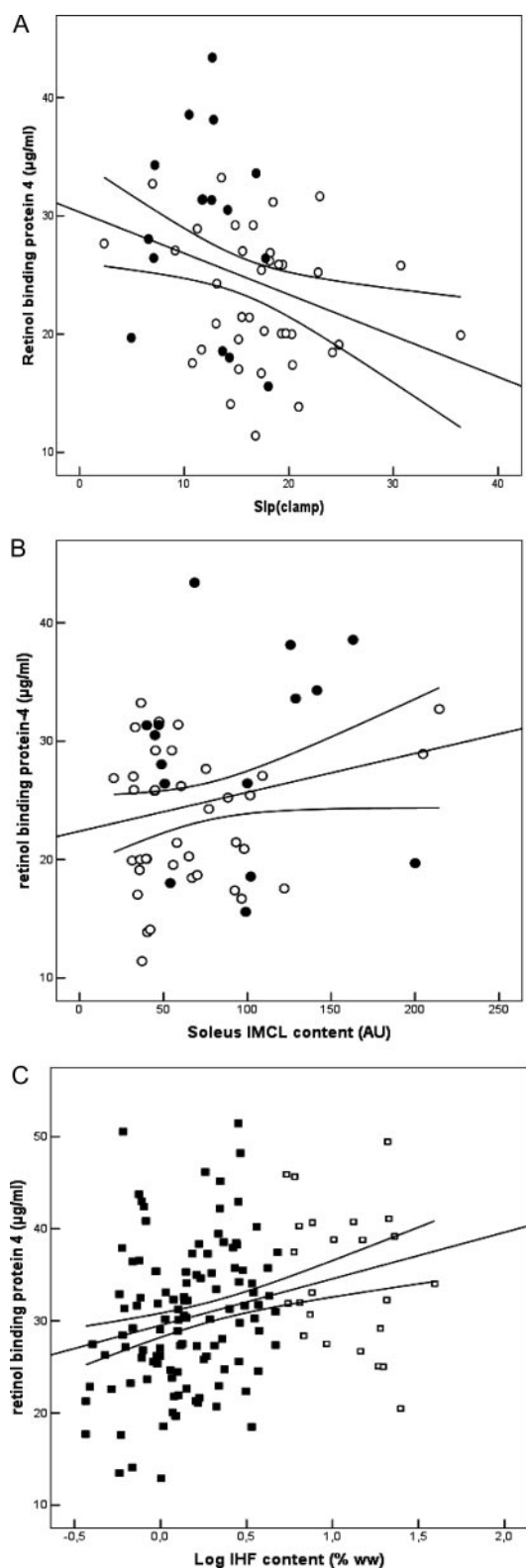


FIG. 1. A, Graphic summary of the correlation between fasting serum RBP-4 concentration and $SI_{P(\text{clamp})}$ (obtained by study 1; ●, offspring of type 2 diabetic parents; ○, normal controls). B, Correlation between the fasting serum RBP-4 concentration and the soleus IMCL content (obtained by study 1; ●, offspring of type 2 diabetic parents; ○, normal controls). C, Correlation between the fasting serum RBP-4 concentration and the IHL content (obtained by study 2; ■, individuals

Results

Methodological consideration

The individuals recruited for studies 1 and 2 were different subjects, and in general, they were not comparable. The cohort of study 1 was younger (27 ± 7 vs. 35 ± 9 yr; $P < 0.01$), and with a higher proportion of women (Pearson χ^2 6.76; $P = 0.009$) whereas the BMI was not different (24.3 ± 6.2 vs. 24.3 ± 3.8 kg/m²; $P = 0.99$) from the cohort of study 2. In addition, they had lower serum insulin (40 ± 18 vs. 73 ± 35 pmol/liter; $P < 0.01$) and RBP-4 (25 ± 7 vs. 31 ± 8 μg/ml; $P < 0.01$) concentrations and higher serum adiponectin (12 ± 6 vs. 8 ± 4 μg/ml; $P < 0.01$) concentration. Plasma glucose (5.1 ± 0.8 vs. 4.8 ± 0.7 mmol/liter; $P = 0.18$) was not different, but individuals recruited in study 1 had higher insulin sensitivity (HOMA2-%S, 112 ± 45 vs. 77 ± 33 %; $P < 0.01$).

Serum adipokines concentration and obesity

Serum RBP-4 was not different by gender and was not associated with age ($r = 0.28$ and $r = 0.14$ in study 1 and 2, respectively; $P > 0.11$) or BMI ($r = -0.13$ and $r = 0.05$ in study 1 and 2, respectively; $P > 0.76$) consistently in both studies 1 and 2. In study 1, no association was found with sex-adjusted percent total body fat ($r = -0.09$; $P = 0.56$), fasting serum FFA ($r = -0.20$; $P = 0.16$), or glycerol ($r = -0.14$; $P = 0.39$) concentrations. Serum adiponectin was not different by gender and was not associated with age ($r = -0.15$ and $r = 0.12$ in studies 1 and 2, respectively; $P > 0.18$). In study 1, it was not associated with BMI ($r = -0.10$; $P = 0.50$), but in study 2, with a wider range of BMI, a significant age- and sex-adjusted correlation was found ($r = -0.27$; $P = 0.01$). No association was found with sex-adjusted percent total body fat ($r = 0.05$; $P = 0.74$), serum FFA ($r = 0.13$; $P = 0.38$), or glycerol ($r = 0.08$; $P = 0.62$) concentrations. Serum leptin was different between gender in both studies ($P < 0.001$) and was consistently associated with age ($P < 0.01$) and BMI ($P < 0.0001$) in both studies 1 and 2 and with serum FFA ($r = 0.20$; $P = 0.05$) and glycerol ($r = 0.27$; $P = 0.02$) concentrations.

Serum adipokines concentration and insulin sensitivity

Serum RBP-4, adiponectin, and leptin were associated with insulin sensitivity [$SI_{P(\text{clamp})}$] in study 1 (Fig. 1A and Table 3). Serum RBP-4 and adiponectin, but not leptin, remained significantly associated when the association was adjusted for age, sex, and body fat. Stepwise regression analysis showed that body fat (step 1) and serum RBP-4 concentration (step 2) independently predicted $SI_{P(\text{clamp})}$ (Table 3). In study 2, serum adiponectin was associated with HOMA2 ($r = -0.18$; $P = 0.044$), whereas RBP-4 showed only a trend ($r = 0.15$; $P = 0.092$), and leptin was not associated ($r = -0.06$; $P = 0.86$) with HOMA2 when adjustment for age, sex, and BMI was taken into account. Stepwise regression analysis did not select any of the above-described adipokines as an independent predictor of HOMA2.

without fatty liver; □, individuals with fatty liver). The measurements in A and B and those in C were obtained in different cohorts. The results of the Pearson correlation analysis, adjustment for age, sex, BMI, or percent body fat, and the stepwise regression analysis are shown in Tables 3 and 4.

TABLE 3. Results of the correlation and multiple regression analysis between $SI_{P(\text{clamp})}$ and the adipokines (study 1)

Variable	r	β	P values
Correlation analysis			
Adiponectin	0.31		0.024
RBP-4	−0.30		0.027
Leptin	−0.30		0.034
Partial correlation analysis (age, sex, and body fat adjusted)			
Adiponectin	0.25		0.031
RBP-4	−0.24		0.041
Leptin	−0.05		0.741
Stepwise multiple regression analysis (including age, sex, body fat, adiponectin, RBP-4, and leptin)			
Step 1			
Body fat	0.39	−0.391	0.000
Step 2			
Body fat + RBP-4	0.49	−0.441 −0.286	0.000 0.044

r represents the correlation coefficient, and β represents the standardized regression coefficient.

Serum adipokines concentration in first-degree relatives of type 2 diabetic parents

A family history of type 2 diabetes was associated with increased serum RBP-4 concentration both in study 1 (Table 1) and study 2 (31.8 ± 7.9 vs. 27.9 ± 6.7 $\mu\text{g/ml}$; $P < 0.01$). Similarly, adiponectin was reduced in both study 1 (Table 1) and study 2 (6.9 ± 2.9 vs. 7.9 ± 3.7 $\mu\text{g/ml}$; $P < 0.05$) in the offspring, whereas no difference was found with respect to serum leptin concentrations.

Serum adipokines concentration and IMCL content

Serum leptin showed a strong correlation with the soleus IMCL content whereas RBP-4 (Fig. 1B), and adiponectin showed a trend for a weak association. Partial correlation showed that also serum RBP-4, but not adiponectin, was significantly associated when adjusting for age, sex, and body fat. Stepwise regression analysis showed that leptin (step 1) and RBP-4 (step 2) concentrations independently predicted the soleus IMCL content (Table 4).

Serum adipokines concentration and IHL content

Serum adiponectin and RBP-4 (Fig. 1C), but not leptin, showed a strong correlation with the IHL content (Table 4). Partial correlation showed that only adiponectin was significantly associated when adjusting for age, sex, and BMI. Stepwise regression analysis showed that BMI (step 1), sex (step 2), and serum adiponectin concentrations (step 3) independently predicted the IHL content (Table 4).

Role of sex

The correlative relationships among adipokines and IMCL content were found to be modulated by sex. In detail, the stepwise regression analysis performed separately between sexes confirmed that the soleus IMCL content was significantly explained by the fasting serum leptin and RBP-4 concentration in women, but this finding was not confirmed in men, in which only serum leptin concentration and age remained significant predictors in the model and RBP-4 was an

TABLE 4. Results of the correlation and multiple regression analysis between soleus IMCL and IHL content and the adipokines

Variable	r	β	P values
Soleus muscle			
Correlation analysis			
Adiponectin	−0.26		0.066
RBP-4	0.22		0.117
Leptin	0.58		0.000
Partial correlation analysis (age, sex, and body fat adjusted)			
Adiponectin	−0.24		0.101
RBP-4	0.29		0.047
Leptin	0.42		0.030
Stepwise multiple regression analysis (including age, sex, body fat, adiponectin, RBP-4, and leptin)			
Step 1			
Leptin	0.57	0.576	0.000
Step 2			
Leptin + RBP-4	0.64	0.615 0.295	0.000 0.011
Liver			
Correlation analysis			
Adiponectin	−0.45		0.000
RBP-4	0.28		0.001
Leptin	0.12		0.104
Partial correlation analysis (age, sex, and BMI adjusted)			
Adiponectin	−0.24		0.005
RBP-4	0.13		0.127
Leptin	0.12		0.187
Stepwise multiple regression analysis (including age, sex, BMI, adiponectin, RBP-4, and leptin)			
Step 1			
BMI	0.64	0.639	0.000
Step 2			
BMI + Sex	0.68	0.525 0.265	0.000 0.000
Step 3			
BMI + Sex + Adiponectin	0.71	0.470 0.280 −0.196	0.000 (0.004) (0.006)

r represents the correlation coefficient, and β represents the standardized regression coefficient.

excluded variable ($P = 0.125$). With respect to the IHL content, we found that in this population, fatty liver was more frequent in men than women (Table 2), confirming our previous report (5). In fact, the stepwise multiple regression analysis (Table 4) selected sex, along with BMI and adiponectin, as an independent variable associated with the IHL content. Despite that, when the stepwise regression analysis was performed separately in each gender category, the IHL content was significantly explained by the BMI and fasting serum adiponectin concentration in both men and women; in addition, serum leptin concentration was also included as a significant variable in both men ($P = 0.043$) and women ($P = 0.05$).

Discussion

Previous work in humans suggested that RBP-4 is associated with whole-body insulin sensitivity. This study performed in nondiabetic subjects confirms that this association

is detectable when explored using clamp-derived indexes of insulin sensitivity (2) and that it may be less apparent when using HOMA-derived indexes (8). The results also emphasize that individuals with a family history of type 2 diabetes are characterized by increased serum concentrations (2). In contrast with previous reports (2, 3, 9), our data suggest that this relationship with insulin resistance is not mediated by obesity because the serum RBP-4 was not associated with parameters of body adiposity. This finding is in agreement with the observation that in a cross-sectional study including 74 menopausal women, RBP-4 gene expression in sc abdominal adipose tissue and RBP-4 serum levels were similar in the normal-weight, overweight, and obese women (10). Also, a recent study by Stefan *et al.* (11) failed to report an association between RBP-4 and total, sc, and visceral amount of fat.

The novel finding of our work is that serum RBP-4, even if not associated with body adiposity, correlated with excessive fat accumulation in nonadipose tissues such as the skeletal muscle and the liver (Table 4). Our data do not allow us to establish the mechanism for the relationship between ectopic fat accumulation and serum RBP-4. Whether RBP-4 is mediating this effect via the control on lipolysis may not be stated based on our data. Using the cross-sectional approach, we failed to find a relationship of this adipokine not only with BMI and body fat content (as assessed by means of dual x-ray absorptiometry) but also with serum fasting FFA and glycerol concentrations.

The effects of adipokines on metabolism and insulin sensitivity are generally studied in isolation (12); understanding the interaction among the adipokines is therefore a mandatory task. An additional aim of the present work was therefore to evaluate the relationship among RBP-4, insulin sensitivity, and ectopic fat accumulation taking also into consideration fasting serum adiponectin and leptin concentrations, also known to be involved in the regulation of insulin sensitivity and body fat distribution. Hence, we compared the association of these adipokines with the parameters of interest applying multiple stepwise regression analysis. Based on this strategy, we learned that 1) the association of RBP-4 with insulin sensitivity was a robust one, because it resulted in being an independent predictor of $SI_{P(\text{clamp})}$ along with body fat mass (Table 3), whereas at the same time, serum adiponectin and leptin were not selected; 2) RBP-4 and leptin were independent predictors of IMCL content, even more robust than body fat content, whereas adiponectin was not; and 3) adiponectin was an independent predictor of the IHL content along with anthropometric features, whereas RBP-4 and leptin were not. In correlative terms, these data strongly agree with a very recent report by Stefan *et al.* (11) using a very similar approach; the only difference is that these authors concluded that the IMCL content was not associated with the RBP-4 based on the negative findings of the univariate analysis (this is similar to our findings reported in Table 4). Using the approach of the stepwise regression analysis, we found that RBP-4 was an independent predictor of the IMCL content.

The strengths of the work were that it was performed in young individuals with a well-characterized phenotype

and no confounding effect of diabetes. In addition, ectopic fat content was assessed noninvasively in a quantitative fashion using ^1H -MRS. Finally, this study did not consider one adipokine, but three adipokines were simultaneously assessed in an attempt to dissect out their relative importance in predicting important metabolic parameters. The work has also obvious limitations. The most important one is its cross-sectional approach; despite that, at least four other studies (2, 9, 11, 13) using a longitudinal approach showed that RBP-4 may be a good marker of insulin resistance in humans. Another limitation is that the measurements of IMCL and IHL contents were obtained in different groups of individuals that were studied in different periods and were not fully comparable for some features as stated in *Results*. The simultaneous assessment of IMCL and IHL content in the same individuals would represent a more robust approach. A methodological work suggested, during the revision of the present manuscript, that the ELISA kit commercially available for the assessment of RBP-4 may be overestimating the circulating levels in the more insulin-sensitive individuals (14); we therefore cannot exclude that the reported RBP-4 associations would have been stronger if more reliable measures (quantitative Western blotting employing full-length RBP-4 protein standards) were employed.

Future investigation is therefore warranted to establish whether a pathogenic link exists among these adipokines, insulin resistance, and ectopic fat accumulation.

In conclusion, these data confirm that fasting serum RBP-4 concentration is associated with insulin resistance and demonstrate that when compared with fasting serum adiponectin and leptin concentrations, RBP-4 was more robust in lean, nondiabetic, insulin-resistant individuals. The sites where these adipokines may potentially be involved in the development of insulin resistance remain uncertain, but their effects may be mediated by ectopic fat accumulation both at the liver and skeletal muscle sites.

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